

SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project - Final Presentation

GitHub repository URL: [INSERT YOUR GITHUB REPOSITORY URL HERE]

Repository should contain the completed notebooks and Python files used in the capstone.

1. Executive Summary

Objective: Predict whether a Falcon 9 first stage will land successfully using launch-related data.

Workflow: SpaceX API + web scraping -> data wrangling -> EDA/SQL -> visualization/Folium/Dash -> classification modeling.

Key course results: 90 Falcon 9 launch records after filtering Falcon 1; SQL exercises identified 55 launches from CCAFS SLC 40, 1 geosynchronous-orbit launch, and 41 successful drone-ship landings.

Outcome: The analysis demonstrates that mission, site, orbit, payload and booster information can be combined to study landing success and support reusable-launch decision making.

2. Introduction and Business Problem

SpaceX advertises Falcon 9 launch services at a lower cost than many traditional providers, partly because the first stage can be recovered and reused.

Business problem: If we can predict first-stage landing success, we can better assess launch economics and competitive launch cost.

Data science question: Can mission characteristics be used to classify a Falcon 9 landing as successful or unsuccessful?

Target: Class = 1 for successful landing and Class = 0 for unsuccessful landing.

3. Data Collection Methodology

API collection: Send GET requests to the SpaceX API, normalize JSON responses with `pandas.json_normalize`, and retain Falcon 9 launch records.

Web scraping: Parse Falcon 9 historical launch tables with BeautifulSoup and extract structured launch attributes.

Collected fields: flight number, date, booster version, payload, orbit, launch site, landing outcome, landing pad and related mission information.

Validation: Inspect dataframe shape, column types, missing values and representative rows before analysis.

4. Data Wrangling Methodology

Remove Falcon 1 launches and focus on Falcon 9 records.

Inspect and handle missing values, especially landingPad and other mission fields.

Create a binary landing-success label from landing outcomes.

One-hot encode categorical features and standardize numeric features before classification.

Split the prepared data into training and test sets; the modeling exercise used 18 test records.

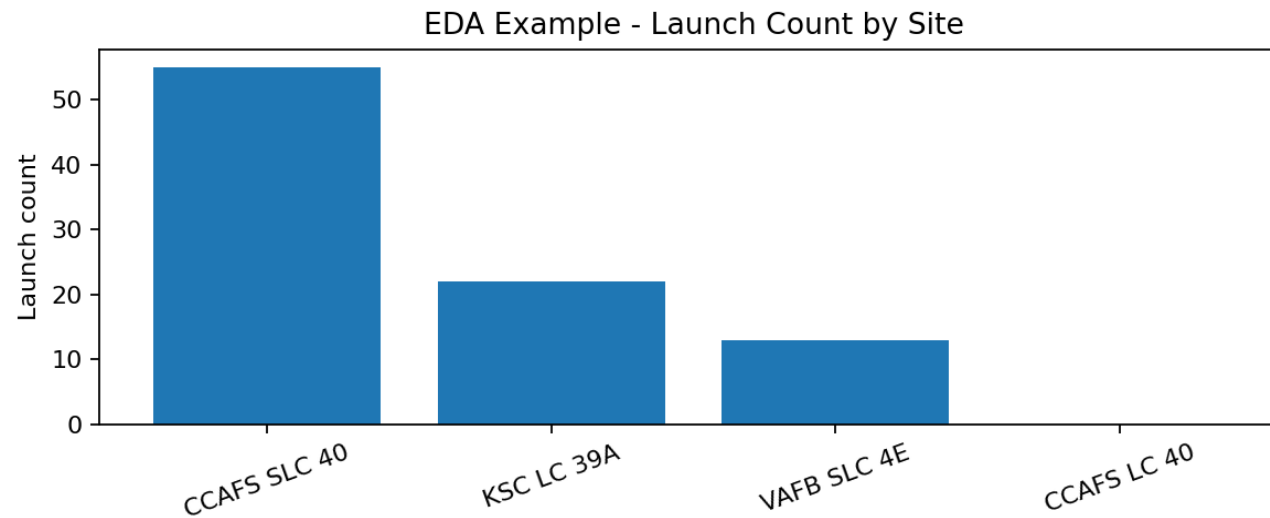
5. EDA and Data Visualization Methodology

Use SQL/Pandas aggregation to count launches and outcomes by site and orbit.

Use scatter plots to study payload mass, flight number and landing success.

Use categorical plots to compare success rates across launch sites and orbit categories.

Use Folium to inspect geographic relationships and Plotly Dash to make the analysis interactive.



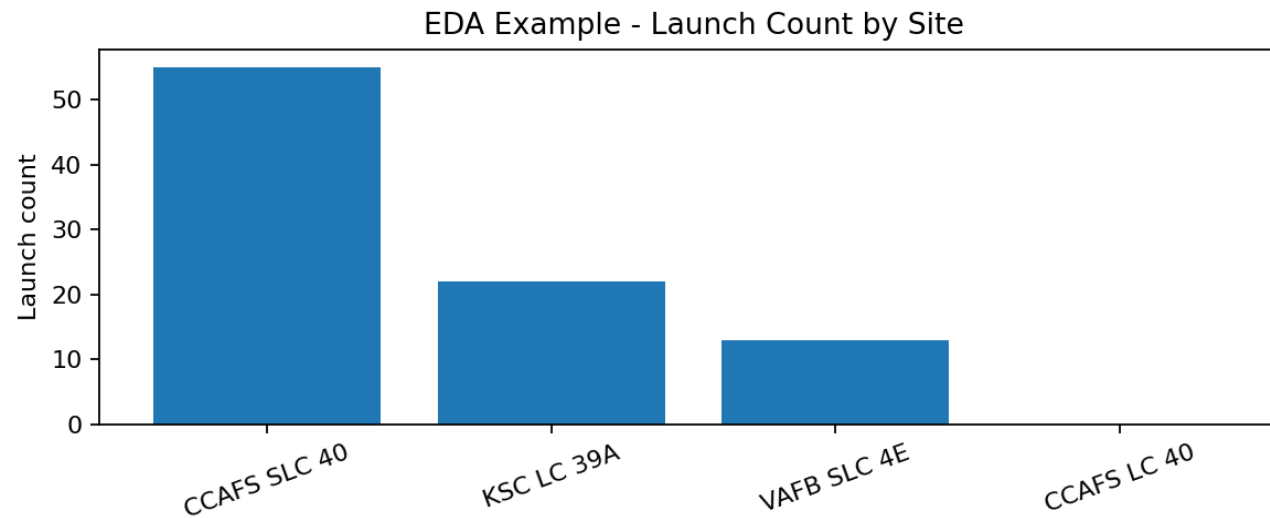
6. EDA with Visualization - Key Observations

Launch activity differs substantially by launch site; CCAFS SLC 40 is the most frequent site in the SQL exercise.

Landing outcomes vary across mission and orbit characteristics, so categorical context matters.

Payload mass and flight sequence can be visualized against landing class to identify clusters and changing success patterns.

These plots motivate the use of multiple features rather than a single-variable rule.



7. EDA with SQL - Queries and Results

SQL was used to answer targeted business questions and validate patterns found during EDA.

SQL analysis question	Coursework result
How many launches came from CCAFS SLC 40?	55
How many launches went to geosynchronous orbit?	1
How many mission outcomes successfully landed on a drone ship?	4
Falcon 9 records after Falcon 1 removal	90

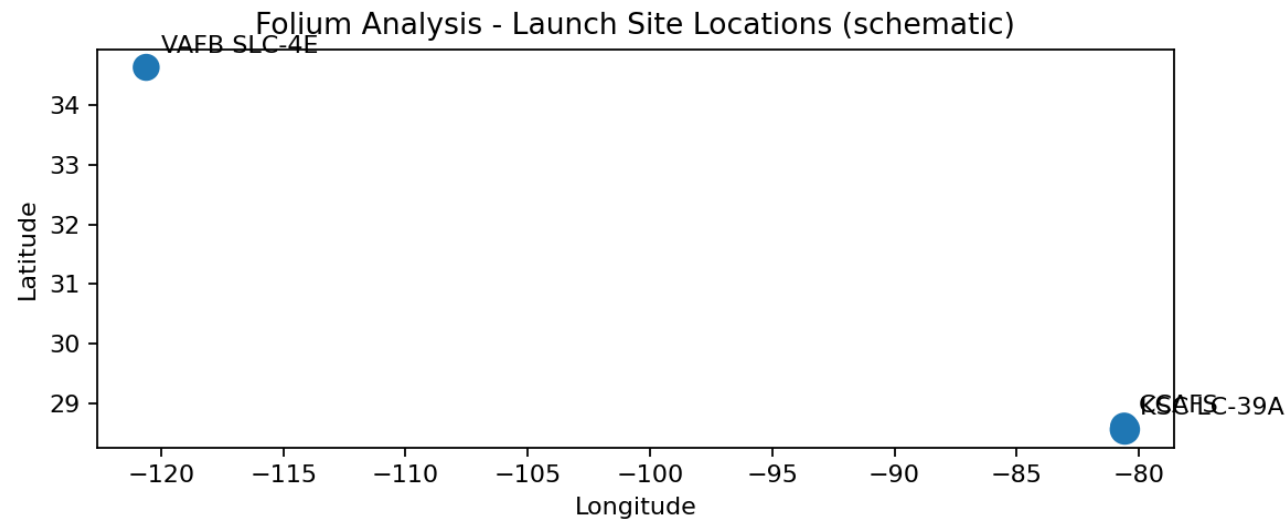
8. Folium Map Analysis

Create an interactive map centered on launch locations and add markers for each launch site.

Use marker/cluster layers and geographic context to inspect proximity to coastlines, roads, railways and populated areas.

The map supports operational interpretation: launch facilities are coastal and geographically separated from dense urban centers.

Evidence expected by rubric: include the actual Folium map screenshot from your notebook if available.



9. Plotly Dash Dashboard

The dashboard provides interactive exploration of launch success.

Controls: launch-site dropdown and payload-range selection.

Visuals: pie chart for success proportions and scatter plot for payload versus outcome/booster category.

Dash callbacks update figures when the user changes an input, enabling rapid comparison among launch sites and payload ranges.

Evidence expected by rubric: replace/add a screenshot of your running dashboard when available.

10. Predictive Analysis - Classification Methodology

Prepare feature matrix X and target Y (landing class), then standardize features.

Split into train/test data and tune candidate models using validation/GridSearchCV.

Compare Logistic Regression, Support Vector Machine, Decision Tree and K-Nearest Neighbors.

Use test accuracy and confusion-matrix interpretation to evaluate generalization.

The course exercise reported 18 records in the test sample.

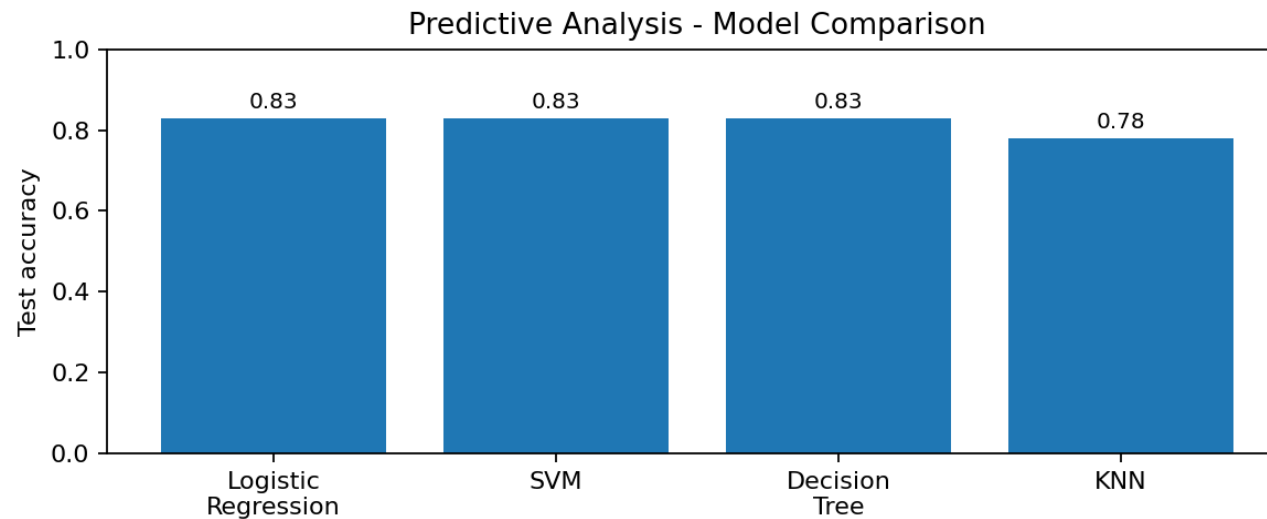
11. Predictive Analysis Results

The SVM validation exercise selected the **sigmoid** kernel among the listed choices.

The Decision Tree exercise produced approximately **83.33% test accuracy** on the first run referenced in the coursework.

Model comparison is necessary because similar validation scores can hide different error patterns.

Final model selection should use the notebook's actual GridSearchCV scores and test results; update the chart if your run differs.



12. Conclusion and Insights

1. Falcon 9 landing success can be analyzed as a supervised binary classification problem.
2. Launch site, orbit, payload and mission history provide useful explanatory context.
3. SQL and visualization reveal operational patterns before modeling.
4. Geospatial analysis adds context about launch-site location and surroundings.
5. Validation and test evaluation are essential; training performance alone is insufficient.

Innovative insight: combining geographic, mission and historical launch features provides a richer view of landing success than analyzing any single variable.

Next step: add newer launch data, weather variables and richer cost data, then retune and re-evaluate the classifiers.

13. Submission Checklist

Required before upload:

- 1) Replace the GitHub placeholder on the title slide with your real repository URL.
- 2) Ensure the repository contains all completed notebooks and Python files.
- 3) If available, replace the schematic EDA/Folium/Dash visuals with screenshots from your actual notebooks/dashboard.
- 4) Confirm every required rubric section appears as a completed slide: Executive Summary, Introduction, Data Collection/Wrangling, EDA methodology, Classification, EDA visualizations, SQL queries, Folium, Dash, Predictive Results and Conclusion.
- 5) Export and upload this final presentation as PDF.